

President of Safety, Health and Sustainability at Tata Steel. Biochar is charcoal made from organic waste.

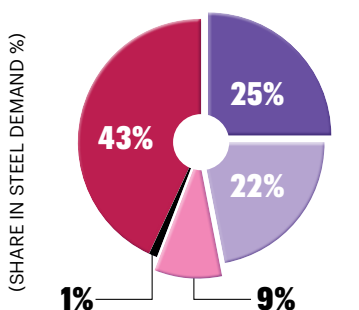
Jindal Steel and Power Ltd, or JSPL, is seeking efficiency improvements and renewable energy integration with investments in solar and wind power generation and green hydrogen, logistics optimisation and ultra-low carbon steel production. "JSPL is aiming for a 50% reduction in coal usage. It is exploring low-carbon technologies along with sustainable logistics through the development of a mechanised Paradip port, deployment of LNG trucks, 80% rail-based steel dispatches, and adoption of electric vehicles for last-mile delivery, in-plant operations and raw material handling," the company said.

Companies are also looking at using green hydrogen, produced by electrolysis of water using renewable energy. The fuel can produce the intense heat required for production of cement and steel. Abhyuday Jindal, Managing Director of Jindal Stainless, says the company plans to invest about ₹700 crore in decarbonisation projects over the next four years. "To lower our carbon footprint, we are expanding our renewable capacity (including captive solar and external PPAs), exploring and expanding the switch to biofuels and piloting production of green hydrogen. In a significant milestone, we recently inaugurated India's first green hydrogen plant in the stainless steel industry," says Jindal. The company is working with the National Institute of Secondary Steel, an arm of the Ministry of Steel, for validation of carbon accounting to gain a green star rating.

Getting such certification is also on the agenda of Vedanta Group's ESL Steel as it prepares to start production of green steel. "We are

STEEL END-USE

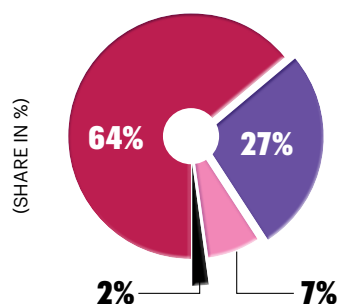
- Construction & infrastructure consume 68% of steel
- Demand will rise as India is yet to build a lot of infrastructure



- BUILDING AND CONSTRUCTION
- INFRASTRUCTURE
- ENGINEERING AND PACKAGING
- AUTOMOTIVE • DEFENCE

SHARE OF CEMENT TYPES

- Most cement in India is blended
- Cement firms are taking steps to reduce emissions



- PORTLAND POZZOLANA CEMENT*
- ORDINARY PORTLAND CEMENT
- PORTLAND SLAG CEMENT*
- COMPOSITE CEMENT*
- * BLENDED; SOURCE INDUSTRY

making measurable progress in reducing our carbon footprint. At the core of our decarbonisation journey is a fundamental shift in how we power our operations and transport systems," says Ravish Sharma, Deputy CEO of Vedanta ESL.

CEMENT MAKERS ACT

India is the world's second-largest producer of cement, contributing 8% of the world's installed capacity and 7% of production.

To decarbonise production, the industry is seeking to enhance energy efficiency through initiatives like waste heat recovery, optimising design to minimise concrete and cement waste, transitioning to ready-mix concrete production and adopting carbon capture technologies.

For instance, in FY24, blended cement accounted for 73.52% of production at Shree Cements, says M.M. Rath, Joint President of Power Plants at the company.

"A key goal is to reduce Net Specific Scope 1 emissions by 12.7% per tonne of cementitious material by 2030, using 2019 as the baseline year. It is being addressed through production efficiency, alternative fuels and integration of cleaner technologies," he says. Net Specific Scope 1 emissions refer to a company's direct greenhouse gas emissions from sources owned and controlled by it.

Kaustubh Phadke, India head of the Global Cement Concrete Association, says over 70% of cement produced in India is blended, using industrial byproducts like fly ash and slag, significantly reducing emissions. "The sector is also leading in co-processing RDF (refuse-derived fuel) derived from municipal solid waste and non-recyclable plastic waste as alternative fuels," says Phadke. **BT**

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